WORKED SOLUTION

STEP II 2017, Question 5

Pure • Parabola Normals and Circles

STEP II 2017, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q5

The key idea

Everything rests on one fact about the parabola $x = at^2$, $y = 2at$: the chord joining the points with parameters t_1 and t_2 has gradient $\dfrac{2}{t_1 + t_2}$. With that, "the normal meets the curve again" and "the angle in a semicircle is a right angle" both become simple equations between parameters, and the geometry turns into short algebra.

Solution

A useful fact: chord gradients

For distinct parameters $t_1 \neq t_2$, the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ has gradient

$$\frac{2a(t_1 - t_2)}{a(t_1^2 - t_2^2)} = \frac{2}{t_1 + t_2},$$

cancelling $t_1 - t_2 \neq 0$. (If $t_1 + t_2 = 0$ the chord is vertical.) Letting $t_2 \to t_1$, the tangent at t has gradient $\frac{1}{t}$; directly, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{2at} = \dfrac{1}{t}$.

Part (i)

At P the tangent has gradient $\frac{1}{p}$, so the normal has gradient $-p$ (here $p \neq 0$ is used). The point N, with $n \neq p$, lies on this normal exactly when the chord PN has gradient $-p$:

$$\frac{2}{p+n} = -p \iff p + n = -\frac{2}{p} \iff n = -\left(p + \frac{2}{p}\right).$$

(The chord PN cannot be vertical, as the normal at P has finite gradient $-p$.) Note $n \neq p$, since $n - p = -2(p + \frac{1}{p})$ and $p + \frac{1}{p} \neq 0$ for real p, so this really is a second point. This is the required result.

Part (ii)

Using $p+n=-\frac{2}{p}$ and $p-n=2p+\frac{2}{p}=\dfrac{2(p^2+1)}{p}$,

$$\begin{aligned}|PN|^2 &= a^2(p^2-n^2)^2+4a^2(p-n)^2\\ &= a^2(p-n)^2[(p+n)^2+4]\\ &= a^2\cdot\frac{4(p^2+1)^2}{p^2}\cdot\left(\frac{4}{p^2}+4\right)=a^2\cdot\frac{4(p^2+1)^2}{p^2}\cdot\frac{4(p^2+1)}{p^2}\\ &= 16a^2\,\frac{(p^2+1)^3}{p^4},\end{aligned}$$

as required.

To minimise, write $u=p^2>0$ and $f(u)=\dfrac{(u+1)^3}{u^2}$. By the quotient rule,

$$f'(u)=\frac{3(u+1)^2u^2-2u(u+1)^3}{u^4}=\frac{(u+1)^2(3u-2u-2)}{u^3}=\frac{(u+1)^2(u-2)}{u^3}.$$

For $u>0$ the factor $\frac{(u+1)^2}{u^3}$ is positive, so $f'(u)<0$ for $0<u<2$ and $f'(u)>0$ for $u>2$. Hence f decreases then increases, and its minimum on $u>0$ is at $u=2$, i.e. $p^2=2$. Since $|PN|^2=16a^2f(p^2)$, this minimises $|PN|$. (The minimum value is $|PN|^2=16a^2\cdot\frac{27}{4}=108a^2$, so $|PN|=6\sqrt{3}\,a$.)

Part (iii)

Since PN is a diameter and Q is a point of the circle other than P and N, the angle PQN is a right angle (angle in a semicircle). So $QP\perp QN$.

? **Ask yourself** Could QP or QN be vertical, so that we can't multiply gradients?

If QP were vertical then QN would be horizontal, so Q and N would have the same y-coordinate, giving $q=n$, which is impossible as $Q\neq N$. Similarly, QN vertical would force QP horizontal and $q=p$. So both gradients are finite and non-zero, and their product is -1:

$$\frac{2}{q+p}\cdot\frac{2}{q+n}=-1\iff(q+p)(q+n)=-4.$$

Expanding, and using $p+n=-\frac{2}{p}$ and $pn=-p^2-2$ from part (i),

$$q^2+(p+n)q+pn=q^2-\frac{2q}{p}-p^2-2=-4,$$

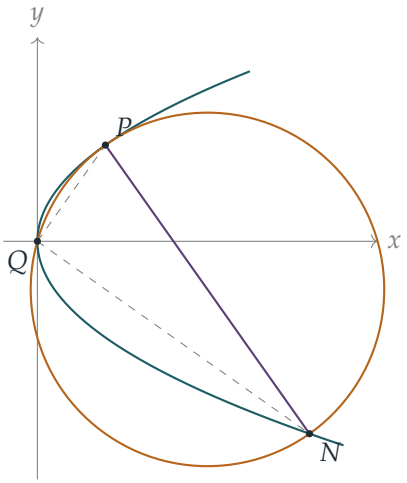
which rearranges to

$$2=p^2-q^2+\frac{2q}{p},$$

as required.

Deduction. Put $q = 0$: the relation becomes $2 = p^2$. Conversely, if $p^2 = 2$ the relation gives $q^2 - \frac{2q}{p} = 0$, i.e. $q = 0$ or $q = \frac{2}{p}$; but $\frac{2}{p} = \frac{p^2}{p} = p$, which is the point P itself, so Q must have $q = 0$. Hence Q is at the origin exactly when $p^2 = 2$, which by part (ii) is exactly when $|PN|$ is at its minimum.

In general, the relation is a quadratic $q^2 - \frac{2}{p}q + (2 - p^2) = 0$ whose roots sum to $\frac{2}{p}$; one root is always $q = p$, so the other is $q = \dfrac{2 - p^2}{p}$, which is zero precisely when $p^2 = 2$.



The figure shows the minimising case $a = 1$, $p = \sqrt{2}$: the normal PN (plum), the circle on PN as diameter (amber), and the right angle at $Q = O$.

Checks

- With $a = 1$, $p = 1$: $n = -3$, so $P = (1, 2)$ and $N = (9, -6)$. Then $|PN|^2 = 64 + 64 = 128$, and the formula gives $16 \cdot 8/1 = 128$.
- With $p = \sqrt{2}$, $a = 1$: $P = (2, 2\sqrt{2})$, $N = (8, -4\sqrt{2})$, and $\overrightarrow{OP} \cdot \overrightarrow{ON} = 16 - 16 = 0$, so the origin does lie on the circle with diameter PN. Here $|PN| = 6\sqrt{3} \approx 10.392$.
- Locating where the circle meets the parabola numerically, for several values of p, gives exactly the parameters n and $\frac{2-p^2}{p}$ (besides P).

Common slips

- Forgetting that $q = p$ also satisfies the final relation, and so wrongly concluding there are two possible positions of Q when $p^2 = 2$.
- Writing the normal gradient as $-\frac{1}{p}$: the tangent gradient is $\frac{1}{p}$, so the normal's is $-p$.

Beyond STEP

A circle and a parabola usually meet in four points, and substituting $x = at^2$, $y = 2at$ into a circle's equation gives a quartic in t with no t^3 term, so the four parameters sum to zero. Here $n + q = -\frac{2}{p} - p + \frac{2}{p} = -p$, so the fourth parameter is p again: the circle touches the parabola at P, as it must, since its radius there lies along the normal. Counting intersections with multiplicity like this is the starting point of algebraic geometry (Bézout's theorem). For more problems in this style, see Siklos, *Advanced Problems in Mathematics*.

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